ENVS 193DS Final

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Set Up

```
# loading packages
library(tidyverse) # general use
library(here) # file organization
library(janitor) # cleaning data frames
library(readxl) # reading excel files
library(scales) # modifying axis labels
library(ggeffects) # getting model predictions
library(MuMIn) # model selection
library(DHARMa) # checking diagnostics

# reading in data
nest <- read_csv(here("data", "nest_data_final.csv"))

# reading in personal data:
mydata <- read_csv(here("data", "stats-personal-data_05_23.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used a generalized linear model to see if native percent cover (water edge) influences probability of American Avocet microhabitat use. In part 2, they used a chi-square test to see if there was a relationship between two categorical variables, bird species (American Avocet, Forster's Tern, Black-necked Stilt) and habitat type (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

b. Suggestions for rewriting

Part 1: Distance to water edge predicts the probability of American Avocet microhabitat use, suggesting that proximity to the water edge influences where American Avocets choose their habitats.

Part 2: The distribution of bird species (American Avocet, Forster's Tern, Black-necked Stilt) differed significantly across habitat (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat), indicating that each species had a preference for habitat.

c. Further analysis

One additional variable that could influence the probability of American Avocet microhabitat use is water temperature measured in degrees Celsius using a temperature probe at each survey point. This variable is continuous and could influence the amount of available prey for American Avocets to forage on. Higher temperatures could lead to less American Avocet habitat use as there would be less prey.

Problem 2. Data analysis

a. Clean and wrangle

```
# creating new object to clean and wrangle data
nests_clean <- nest |>
  # filtering to only include ant species C. mimosae, C. sjostedti,
  # and C. nigriceps
   filter(ant_species %in% c("RRB", "AB", "BBR")) |>
  # creating new column with full scientific name
  mutate(species_name = recode_values(
    ant_species,
    "RRB" ~ "Crematogaster mimosae",
    "AB" ~ "Crematogaster sjostedti",
    "BBR" ~ "Crematogaster nigriceps"
    )) |>
  # reordering ant species
  mutate(species_name = as_factor(species_name),
         species_name = fct_relevel(species_name,
                                    "Crematogaster sjostedti",
                                    "Crematogaster mimosae",
                                    "Crematogaster nigriceps")) |>
# selecting only columns representing tree height,
# if a bird nest is in tree, ant species, and ant species name
    select(height_cm, case_control,
      species_name, ant_species)
# checking structure of data frame
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ height_cm   : num [1:270] 392 310 513 154 324 ...
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ species_name: Factor w/ 3 levels "Crematogaster sjostedti",..: 2 2 2 2 1 2 3 2 1 3 ...
 $ ant_species : chr [1:270] "RRB" "RRB" "RRB" "RRB" ...
```

```r
# displaying nests clean data fram
slice_sample(nests_clean,
             n = 10) # 10 rows
```

```
# A tibble: 10 x 4
   height_cm case_control species_name            ant_species
       <dbl>        <dbl> <fct>                   <chr>
 1      420.            1 Crematogaster mimosae   RRB
 2      131             0 Crematogaster mimosae   RRB
 3      249             0 Crematogaster mimosae   RRB
 4      130             0 Crematogaster mimosae   RRB
 5      332.            0 Crematogaster mimosae   RRB
 6      176             0 Crematogaster mimosae   RRB
 7      243             0 Crematogaster mimosae   RRB
 8      158.            1 Crematogaster nigriceps BBR
 9      524.            1 Crematogaster mimosae   RRB
10      142             0 Crematogaster mimosae   RRB
```

b. Response variable

The 1s in the column case control means that the tree contained a nest and the 0s mean that there was an "available" for the birds to nest in. The researchers identified available trees by identifying the four nearest neighboring trees above 0.5 m tall to trees with nests.

c. Purpose of study

Tree height

Whistling thorn tree height, measured in centimeters, is a continuous variable. The probability of bird nest occurrence would likely increase as tree height increases since there higher trees would be less accessible to ground-level predators. This information was found in paragraphs 3 and 4 of the introduction section.

Acacia Ant species

Acacia ant species is a categorical variable with three different leveles in our model (*C. mimosea*, *C. nigriceps*, and *C. sjostedti*). Each ant species provides different benefits to the tree and ecosystem. Ant species, like *C. mimosae* and *C. nigriceps*, that have higher aggression compared to other acacia ant species will probably increase the probability of nest occurrence as there are less large predators for the bird nests. This information was found in paragraphs 2 and 3 in the introduction section.

d. Table of models

Model Number	Tree Height	Ant Species	Model Description
1	X	X	all predictors (no interaction)
2	X	X	all predictors (interaction term)

e. Run the models

```
# model 1: no interaction, case control as a function of tree height and
# ant species
model1 <- glm(
  case_control ~ height_cm + species_name,
  data = nests_clean, # data frame: nests clean
  family = binomial) # binomial data

# model 2: interaction term, case control as a function of tree height and ant
# species
model2 <- glm(
  case_control ~ height_cm * species_name,
  data = nests_clean, # data frame: nests clean
  family = binomial) # binomial data
```

f. Select the best model

```
# using Akaike's Information Criterion (AIC) to choose best model
AICc(
  model1, # model 1
  model2 # model 2
```

```r
) |>
  # arange by decreasing model
  arrange(AICc)
```

```
       df     AICc
model2  6 238.1236
model1  4 258.8940
```

The best model as determined by Akaike's Information Criterion (AIC) is the model with the interaction term between ant species and tree height as it explains the most variation in the nest occurrence in the simplest format.

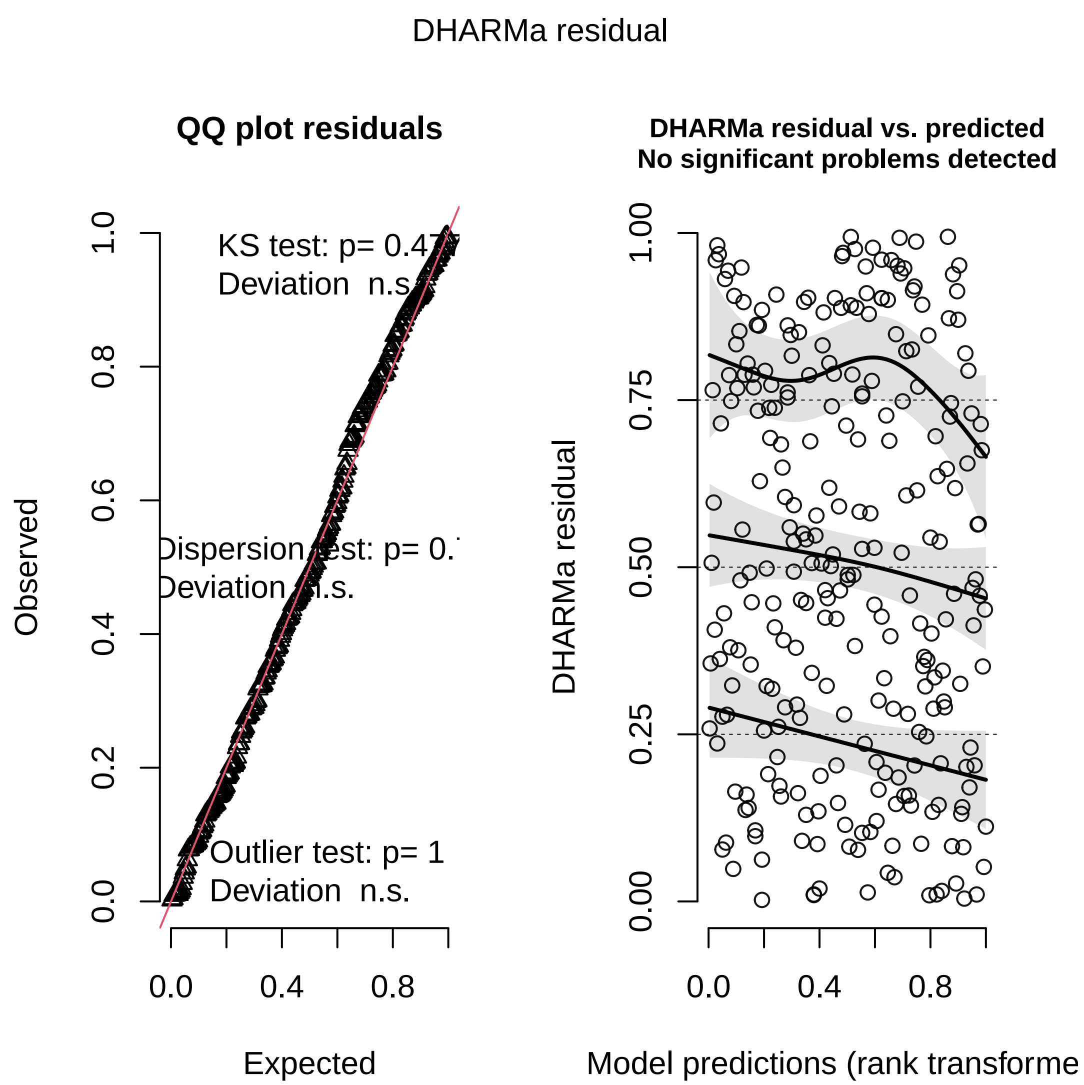
```r
# displaying summary for best model
summary(model2)
```

```

Call:
glm(formula = case_control ~ height_cm * species_name, family = binomial,
    data = nests_clean)

Coefficients:
                                                Estimate Std. Error z value
(Intercept)                                    -3.693410   2.517395  -1.467
height_cm                                       0.003895   0.008237   0.473
species_nameCrematogaster mimosae               0.848334   2.554617   0.332
species_nameCrematogaster nigriceps           -12.279738   5.946723  -2.065
height_cm:species_nameCrematogaster mimosae     0.002597   0.008386   0.310
height_cm:species_nameCrematogaster nigriceps   0.084541   0.031961   2.645
                                              Pr(>|z|)
(Intercept)                                    0.14233
height_cm                                      0.63634
species_nameCrematogaster mimosae              0.73983
species_nameCrematogaster nigriceps            0.03893 *
height_cm:species_nameCrematogaster mimosae    0.75679
height_cm:species_nameCrematogaster nigriceps  0.00817 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 286.04  on 269  degrees of freedom
```

```
Residual deviance: 225.80  on 264  degrees of freedom
AIC: 237.8

Number of Fisher Scoring iterations: 7
```

g. Check the diagnostics

```r
# checking diagnostics for model using DHARMa package
plot(simulateResiduals(model2))
```



From the residuals, the data is uniformly distributed as the plot follows the reference line and outliers do not have significant influence on the data since the lines follow the bands and are relatively flat.

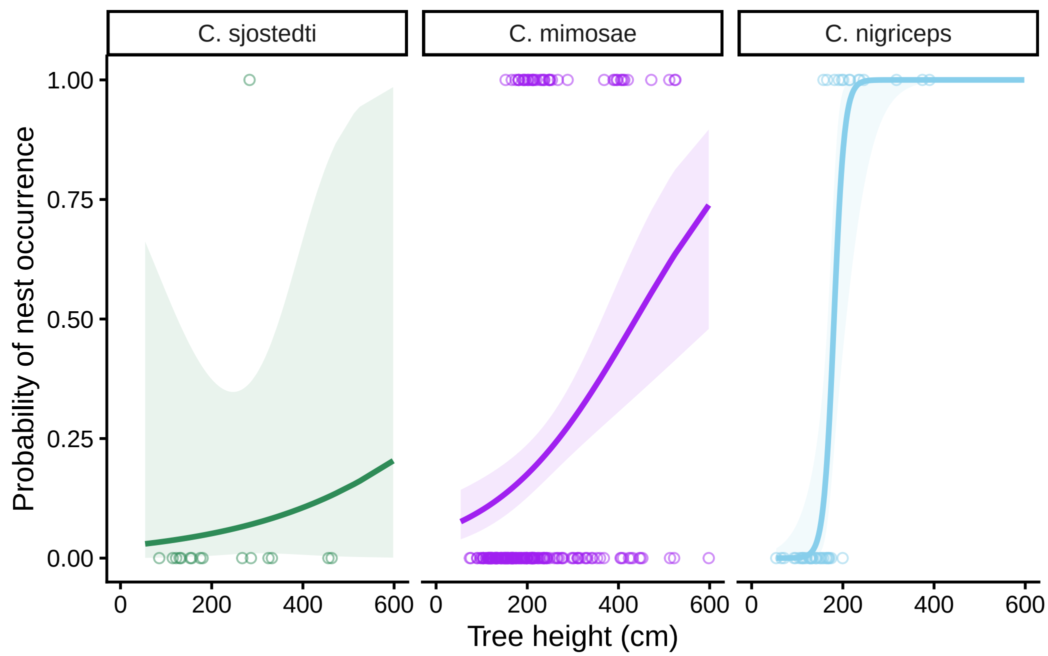
h. Visualize the model predictions

```
# creating new object for predictions
preds <- ggpredict(
```

```
  model2, # using correct model
  terms = c("height_cm[all]", "species_name")) |> # predictors
  # renaming columns to fit data
  rename(height_cm = x,
         species_name = group)
```

```
# base layer: ggplot using cleaned nest ddata
ggplot(data = nests_clean,
       mapping = aes(x = height_cm, # x- axis: tree height
                     y = case_control )) + # y axis: nest occurrence
  # first layer: points
  geom_point(aes(color = species_name), # coloring by ant species
             shape = 21, # open circles
             alpha = 0.5) + # making circles transparent
  # second layer: line using model predictions
  geom_line(data = preds,
            mapping = aes(x = height_cm, # x-axis: tree height
                          y = predicted, # y-axis: probability of nest
# occurrence
                          color = species_name), # categorical variable:
# ant species
            linewidth = 1) + # making line thicker
  # third layer: model predictions with 95% CI
  geom_ribbon(data = preds,
              mapping = aes(
                        y = predicted,# y- axis: probability of nest occurrence
                        ymin = conf.low, # upper 95% CI
                        ymax = conf.high, # lower 95% CI
                        fill = species_name), # filling by ant species
               alpha = 0.1) + # making transparent
  # creating different panels for each ant species
  facet_wrap(~ species_name,
             # adding headers of ant species
             labeller = labeller(
             species_name = c(
                 "Crematogaster sjostedti" = "C. sjostedti",
                 "Crematogaster mimosae"   = "C. mimosae",
                 "Crematogaster nigriceps" = "C. nigriceps"))) +
  # creating custom colors of lines, ribbons, and points
  scale_color_manual(values = c(
    "Crematogaster sjostedti" = "seagreen4",
    "Crematogaster mimosae" = "purple",
```

```r
    "Crematogaster nigriceps" = "skyblue")) +
  scale_fill_manual(values = c(
    "Crematogaster sjostedti" = "seagreen4",
    "Crematogaster mimosae"   = "purple",
    "Crematogaster nigriceps" = "skyblue")) +
scale_x_continuous(limits = c(0, 600), # set x axis to full range of tree height
                   breaks = c(0, 200, 400, 600)) + # even breaks from 600 cm
# relabelling x- and y- axes
labs (x = "Tree height (cm)",
      y = "Probability of nest occurrence") +
# changed theme from default
theme_classic() +
# removed legend
theme(legend.position = "none") +
# removing grid lines and grey background
theme(panel.grid = element_blank(),
      panel.background = element_blank())
```



i. Write a caption for your figure

Figure 1. **Probability of Nest Occurrence increases as tree height increases across all three ant species.** Points represent individual observations of nest occurrence (0 =

no nest, 1 = nest present), with different colors representing different ant () species. Lines represent predicted probability of nest occurrence using a binomial generalized linear model. Shaded ribbon represents 95% confidence interval around model predictions for each species. Data citation: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. https://doi.org/10.5281/zenodo.8373322

j. Calculate model predictions

```
# calculating predicted probability of nest occurence at a value of 600 cm
ggpredict(model2,
          terms = c("height_cm [600]",
                    "species_name"))
```

```
# Predicted probabilities of case_control

species_name: Crematogaster sjostedti

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.20 | 0.00, 0.99

species_name: Crematogaster mimosae

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.74 | 0.48, 0.90

species_name: Crematogaster nigriceps

height_cm | Predicted |     95% CI
----------------------------------
      600 |      1.00 | 1.00, 1.00
```

k. Interpret your results

Ant species and tree height interact to predict probability of nest occurrence. As Whistling thorn tree height increases, the probability of nest occurrence also increases across all three ant species, this effect differs depending on ant species. At the tallest tree height in the dataset, 600 cm, the predicted probability for nest occurrence was highest for *C. nigricps* (1.00, 95% CI: [1.00,1.00]), followed by *C. mimosae* (0.74, 95% CI: [0.48, 0.90]), and lowest for *C. ajostedti* (0.20, 95% CI:[0.00, 0.99]). As seen in Figure 1, the steepest increase in nest probability with tree height occurs in trees with *C. nigricepts*. Biologically, this means that birds are more likley to select taller trees occupied by agressive ant species like *C. nigriceps* and *C. mimosae* for nesting as they can deter large herbivores from the tree.

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

- For the affective visualization, I included all the different variables regarding my data versus the exploratory visualizations isolated two variables. Additionally, for the explanatory visualization, I displayed median and IQR but that was not displayed on my affective visualization.
- Both of my visualizations utilize color to differentiate categorical variables such as work location. Additionally, both visualizations represent all 30 observations, just represented differently.
- The main pattern that is visible in both visualizations is that when I worked at school, I had a higher productivity level than when I worked at any other place like the cafe or home. This pattern is the same throughout visualizations as you can see there's more z's in the affective visualization and higher median on the explanatory visualization.
- The main feedback I got in workshop was to make the variables more clear and be more clear with the overarching message of the data. I implemented this feedback by making the cloud size dependent on productivity level, so it was clear to see if there was a relationship between my variables and productivity levels. The decision to kept the suggestion was based on the fact that it helped convey the message behind the results better and was more clear for the reader.

b. Designing an analysis

The research question I asked was *Does the amount of sleep I get affect my productivity?* and so I will be performing a Spearman's rank correlation test to determine if there is a relationship between sleep and productivity levels to answer if sleep is related to productivity. Both of these variables are continuous and independent of each other. Given that the there is not a linear relationship between variables and a small sample size, a non-parametric version of a correlation test is appropriate compared to a parametric version.

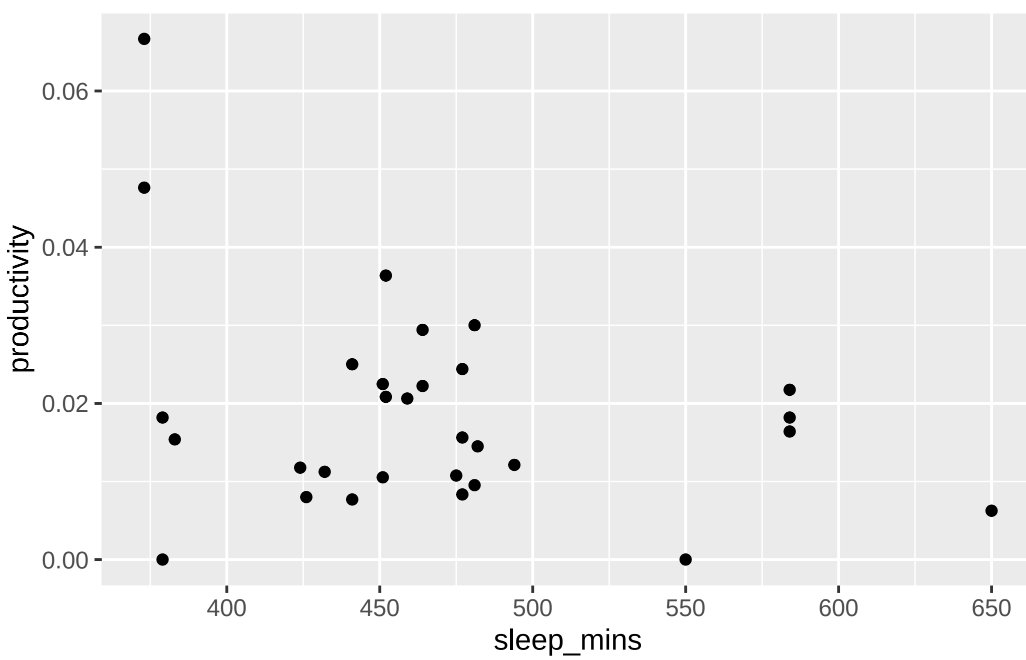
c. Check any assumptions and run your analysis

```
# creating new object from my_data
mydata_clean <- mydata |>
  # cleaning names
  clean_names() |>
  # dropping na values
  drop_na() |>
  # making productivity larger numbers, ), convert column to numeric,
  # instead of character
  mutate(productivity = as.numeric(productivity),
         productivity_hun = productivity*100)
```

Checking assumptions

Check 1: monotonic relationship between variables

```
ggplot(data = mydata_clean,
       mapping = aes(x = sleep_mins,
                     y = productivity)) +
  geom_point()
```



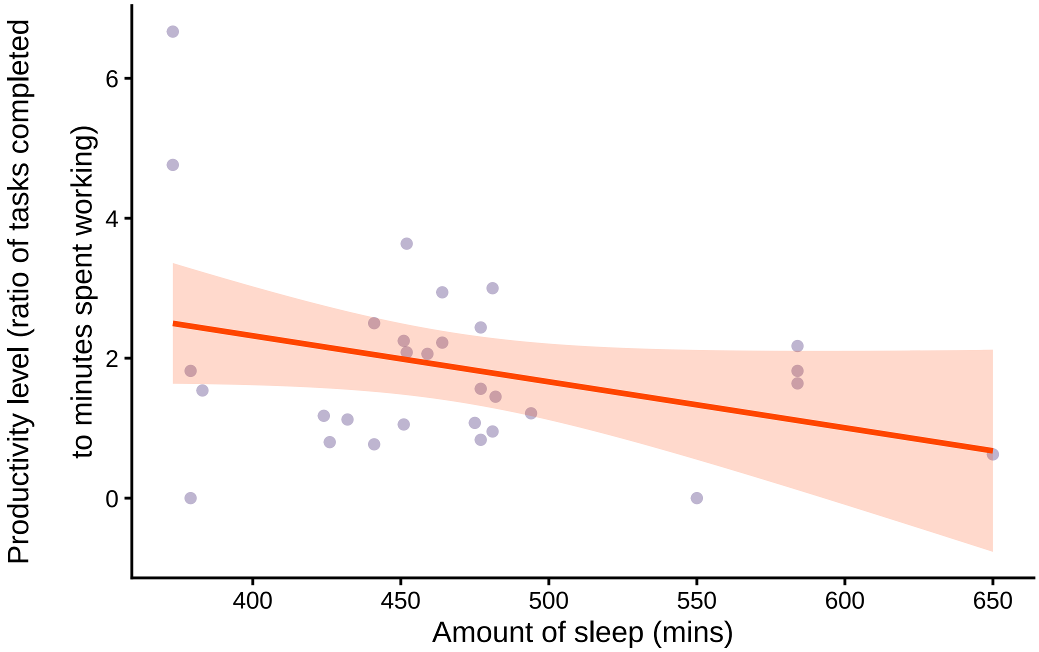
Running analysis

```
# running correlation test
cor.test(mydata_clean$productivity_hun, mydata_clean$sleep_mins, # variables
# productivity levels and sleep in mins
         method = "spearman") # running Spearman's rank  based on assumptions
```

```
	Spearman's rank correlation rho

data:  mydata_clean$productivity_hun and mydata_clean$sleep_mins
S = 5197.3, p-value = 0.4096
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.1562467
```

d. Create a visualization

```
# base layer: ggplot using clean personal data
ggplot(mydata_clean,
       mapping = aes(x = sleep_mins, # x axis: amonunt of sleep
                     y = productivity_hun)) + # y- axis: productivity level
  # first layer: points to show observations
  geom_point(color = "mediumpurple4", # making points
             alpha = 0.4) + # adjusting transparency
  # second layer: showing trend
  geom_smooth(method = "lm",
              color = "orangered",
              fill = "orangered",
              alpha = 0.2) +
  labs(x = "Amount of sleep (mins)",
       y = "Productivity level (ratio of tasks completed
       \nto minutes spent working)") +
  theme_classic()
```



e. Write a caption

f. Write about your results

There was no statistically significant relationship between amount of sleep and productivity level (Spearman's rank, ρ = -0.15, S = 5197.3, p = 0.41, α = 0.05). Although correlation coefficient (ρ = -0.15) suggests a weak negative relationship, the large p-value suggests that this difference statistically different from chance which could be due to small sample size and high variability in the data.